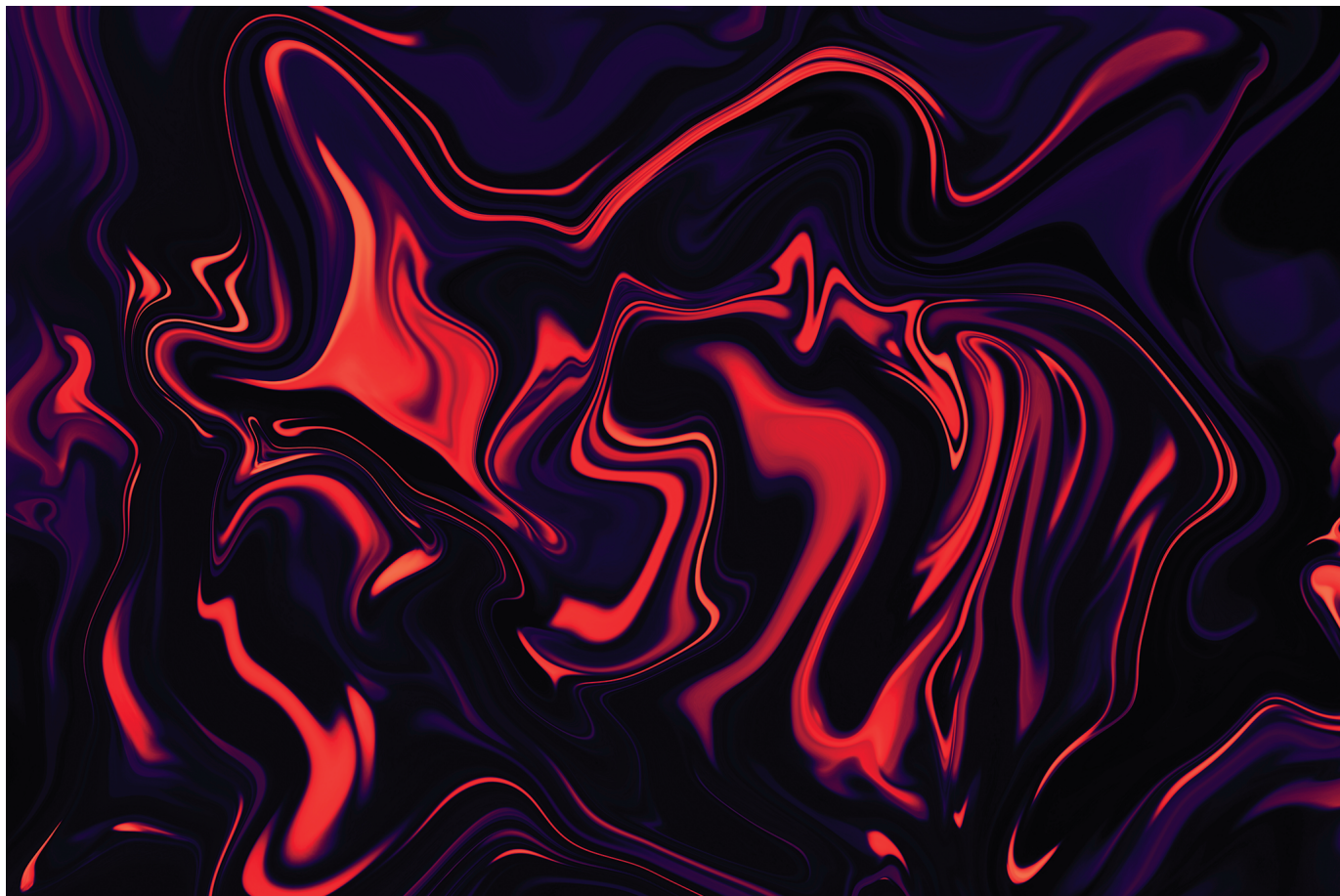

In Search of the Viscosity Operator on Riemannian Manifolds



Magdalena Czubak

Introduction

What should equations modeling fluid flow on Riemannian manifolds look like? As we will see, there are several candidates, and there is no generally agreed upon consensus. We explore this question from the point of view of geometry, analysis, probability, and physics.

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There are two well-studied fundamental equations of fluid flow. They are the Euler equations and the Navier–Stokes equations. The main difference between the Euler equations and the Navier–Stokes equations is that the Navier–Stokes equations incorporate what is called *viscosity*. We explain this difference more below. What is important to note now is that there is no ambiguity as to what the Euler equations should be on a general Riemannian manifold. What is less clear is what the Navier–Stokes equations should be. This lack of ambiguity for the Euler equations will become clear as well.

To motivate this, to see that it might not be completely obvious what the Navier–Stokes equations on a manifold are, we start with the Navier–Stokes equations in the Euclidean setting, and see if we can write the equations on a general manifold.

We begin with an incompressible Navier–Stokes equation on $\mathbb{R}^n$. It is given by

$$\begin{aligned}\partial_t u - \Delta u + u \cdot \nabla u + \nabla p &= 0, \\ \operatorname{div} u &= 0.\end{aligned}\tag{1}$$

We explain now the notation and the unknowns. The unknowns are u and p , where

$$\begin{aligned}u &: \mathbb{R}^{n+1} \rightarrow \mathbb{R}^n, \\ p &: \mathbb{R}^{n+1} \rightarrow \mathbb{R}.\end{aligned}$$

So u is a time-dependent vector field, and p is a scalar-valued function. Physically, $u(x, t) = (u^1(x, t), \dots, u^n(x, t))$ is the velocity of the fluid going through position x at time t , and p is the pressure. The notation in (1) means the following:

- $\partial_t = \frac{\partial}{\partial t}$.
- Δ is the Laplacian:

$$\Delta = \sum_{i=1}^n \partial_i^2,\tag{2}$$

where $\partial_i = \partial_{x^i} = \frac{\partial}{\partial x^i}$.

- ∇p is the gradient of p : $\nabla p = (\partial_1 p, \dots, \partial_n p)$.

Note that since ∇p is the vector, and it appears in the equation, this means that all the other terms are also vector valued, so $\partial_t u$ is interpreted as taking the time derivative of each component u^i , and similarly, Δu means we apply the Laplacian to each u^i .

- $u \cdot \nabla u$ is written here how it is often denoted, but it is more precise to write it as a dot product of u with the gradient of each component u^i , $(u \cdot \nabla u)^i = u \cdot \nabla u^i$, or we can think of this term as the directional derivative of u in the direction of u , so $u \cdot \nabla u = \nabla_u u$.
- Finally, $\operatorname{div} u$ is the divergence of u , $\operatorname{div} u = \partial_i u^i$, where we sum over the repeated indices.

The condition that $\operatorname{div} u = 0$ means that the fluid is incompressible. This is an equivalent condition of saying that it is volume preserving: if we take the fluid in any region, measure its volume, and follow it, then at a later time it will have the same volume. We note that incompressibility is an approximation: there is no fluid that is truly incompressible, even water, but it is a *good* approximation (see, e.g., [Bat99]).

Since the equation is time dependent, we also prescribe initial conditions, and give the fluid the initial velocity: $u(0, x) = u_0(x)$.

Finally, we remark, that while we have two unknowns (u, p) , the first equation in (1) is an evolution equation

only for the velocity vector field u . As a result, one usually solves for u first, and then later the pressure p is recovered from u . For example, we can take the divergence of the first equation in (1) and arrive at an elliptic equation for p .

So now, given (1), the most straightforward way to begin to generalize the system to the Riemannian manifolds is to look at each term and find its analog on a manifold.

All the terms, except for one, have a clear analog. Which one does not? First, there is a natural concept of divergence, of gradient, of directional derivative, and since the manifold is fixed in time (we are not looking at relativistic fluids, but see below), we can also just take the time derivative. The only term that is not obvious is the Laplacian.

Different Laplacians

When we hear the words manifold and Laplacian, there is a good chance that the first operator that comes to mind is the Laplace–Beltrami operator, the divergence of the gradient,

$$\operatorname{div} \nabla,$$

which we can write in coordinates as

$$\frac{1}{\sqrt{\det g}} \partial_{x^i} (\sqrt{\det g} g^{ij} \partial_{x^j}),\tag{3}$$

where $\det g$ denotes the determinant of the metric $g = (g_{ij})$ written in local coordinates $\{x^i\}$, (g^{ij}) is the metric's inverse, and where we sum over the repeated indices.

For example, if the manifold is the Euclidean space, $\mathbb{R}^n$, then in the Cartesian coordinates, the metric is $g_{ij} = \delta_{ij}$, where δ_{ij} is the Kronecker's delta. Then, $\det g = 1$, $g^{ij} = \delta^{ij}$, and the operator reduces to

$$\partial_{x^i} (\delta^{ij} \partial_{x^j}) = \delta^{ij} \partial_{x^i} \partial_{x^j} = \sum_{i=1}^n \partial_{x^i}^2,$$

which is the Laplacian in (2).

However, the operator in (3) acts on scalar-valued functions, but a solution of the Navier–Stokes equation is a vector field, so this operator cannot be used here. We elaborate this point. It is true that when we discussed the Laplacian in (1) we said we apply (2) to each coordinate. In general, in geometry we look for operators that are independent of the choice of coordinates. We could show that even in $\mathbb{R}^2$, if we wrote the vector field in polar coordinates and applied the scalar Laplacian to each coordinate, we would not arrive at an equivalent expression to the one in Cartesian coordinates. So in the introduction, that was the magic of the Cartesian coordinates: they are the most straightforward way to introduce the Navier–Stokes equations on $\mathbb{R}^n$, and allow us to write the Laplacian of the vector field by applying (2) coordinate-wise. In general, we need a Laplacian operator that acts on vector fields and has an expression that is independent of coordinates.

There are actually several candidates for the choice of the Laplacian that acts on vector fields.

To begin with, if we like to think about the Laplacian as the divergence of the gradient, then the natural generalization of the Laplace–Beltrami operator to vector fields is the Bochner Laplacian (sometimes also still called the Laplace–Beltrami operator)

$$\operatorname{div} \nabla,$$

where now ∇ denotes the Levi-Civita connection on (M, g) . Recall, the Levi-Civita connection is an operator that allows us to differentiate vector fields on a manifold. It is also a unique operator that satisfies certain properties (compatibility with the metric and the torsion free property).

In local coordinates, we can write

$$\operatorname{div} \nabla = \nabla^i \nabla_i = g^{ij} \nabla_j \nabla_i,$$

which we observe can also be thought of as taking the trace of ∇^2 , which is another name for the Bochner Laplacian: the trace Laplacian, or the rough Laplacian (there might be sign conventions involved, too).

To see why this is a natural generalization of (3), we remark that the Levi-Civita connection induces an operator that acts on general tensors, and in particular, on 0-rank tensors, which are just functions. In that case, we could show that the induced operator is a differential of a function, d , and then a computation would show that $\operatorname{div} d$ is exactly (3).

Next, when v is a vector field on $\mathbb{R}^3$, the Laplacian of v can be expressed as

$$(\nabla \operatorname{div} - \operatorname{curl} \operatorname{curl})v. \quad (4)$$

Now, when v is a vector field on a Riemannian manifold, then using the metric, we can lower the index, and obtain a unique 1-form α associated to v . In local coordinates, if $v = v^i \partial_{x^i}$, then $\alpha = g_{ij} v^j dx^i$.

More precisely, we can use the so-called musical isomorphisms to move between vector fields and 1-forms (raise/lower indices). If v is a vector field, then $v^\flat$ is a 1-form, and if α is a 1-form, $\alpha^\sharp$ is a vector field. We have the following definitions, which rely on the duality of 1-forms and vector fields: 1-forms act on vector fields.

$$v^\flat = g(v, \cdot), \quad \alpha(\cdot) = g(\alpha^\sharp, \cdot).$$

Then for a 1-form α , $\alpha = v^\flat$, the analog of (4) is

$$-(d d^* + d^* d)\alpha,$$

which is the Hodge Laplacian acting on α . Here, d^* is the formal adjoint of the exterior differential operator d with respect to the inner product given by the metric. The Hodge Laplacian acts on differential forms, and it can also act on functions, and interestingly enough, in that case, just like the Bochner Laplacian, it reduces to (3).

We introduced the Bochner Laplacian as acting on vector fields, but using the musical isomorphisms, we could also have it act on 1-forms, or we could use the musical isomorphisms to define the Hodge Laplacian as acting on vector fields. Either way, we can relate the Bochner Laplacian and the Hodge Laplacian by what is called the Bochner–Weitzenböck formula. The formula is a direct computation using Ricci identities, and it reads

$$-\operatorname{div} \nabla = d d^* + d^* d - \operatorname{Ric}, \quad (5)$$

where Ric is the Ricci operator obtained from the Ricci tensor by raising an index, and more precisely, if α is a 1-form

$$\operatorname{Ric} \alpha = \operatorname{Ric}(\alpha^\sharp, \cdot).$$

It follows that, in the particular case of $\mathbb{R}^n$, where $\operatorname{Ric} \equiv 0$, we can see that the Bochner Laplacian and the Hodge Laplacian agree. On a manifold that has a constant sectional curvature, λ ,

$$\operatorname{Ric} \alpha = (n-1)\lambda \alpha.$$

In the case of the sphere, S^n ,

$$\operatorname{Ric} \alpha = (n-1)\alpha,$$

and on a hyperbolic space, $\mathbb{H}^n$,

$$\operatorname{Ric} \alpha = -(n-1)\alpha.$$

There is another operator that can be considered on a Riemannian manifold. The first in-depth study of the Navier–Stokes equations on the Riemannian manifolds was presented by Ebin and Marsden in their seminal 1970 article [EM70]. In their article, Ebin and Marsden indicated that when considering the Navier–Stokes equations on an Einstein manifold, one should use the following operator

$$2 \operatorname{div} \operatorname{Def},$$

where Def is the deformation tensor. The deformation tensor can be thought of as a symmetrization of the covariant derivative, and in coordinates we can write it as

$$(\operatorname{Def} v)_{ij} = \frac{1}{2}(\nabla_i v_j + \nabla_j v_i). \quad (6)$$

A direct computation using (5), and a Ricci identity gives

$$\begin{aligned} -2 \operatorname{div} \operatorname{Def} &= -\operatorname{div} \nabla + d d^* - \operatorname{Ric} \\ &= d d^* + d^* d + d d^* - 2 \operatorname{Ric}. \end{aligned}$$

For divergence-free vector fields, we can cancel out $d d^*$ terms and just write

$$-2 \operatorname{div} \operatorname{Def} = -\operatorname{div} \nabla - \operatorname{Ric} = d^* d - 2 \operatorname{Ric}. \quad (7)$$

This formula connects all the three operators, and again we can see that they are all the same in case of the Euclidean space.

The article [EM70] states that the “correct” viscosity operator on a manifold should be the operator given by (7), but the article itself uses the Hodge Laplacian, and the

deformation tensor is mentioned only at the end, in the “Note Added in Proof.”

Since 1970 we see mostly the use of either the Hodge Laplacian or the Laplacian coming from the deformation tensor, operator (7). This operator has been called the Ebin–Marsden Laplacian, or just the deformation Laplacian.

Authors working with the Hodge Laplacian are usually working on a compact manifold, including Ebin and Marsden [EM70], so a heuristic notion might be that the Ricci term might not make a difference. However, this is not true, even in the case of compact manifolds as we will discuss when we mention the work in [ST20].

In the rest of this article, we will talk about different perspectives and arguments that we can use to help us decide which operator to work with.

Continuum Mechanics: Why the Deformation Tensor

As mathematicians, we are used to seeing the equation as written in (1). That way, we first write the linear term, the heat operator, then the nonlinearity and the pressure. Engineers, on the other hand, often write the equation as

$$\partial_t u + u \cdot \nabla u = -\nabla p + \mu \Delta u, \quad (8)$$

or more generally as

$$\partial_t v + v \cdot \nabla v = f + \operatorname{div} T. \quad (9)$$

The equation (9) emphasizes the parts written in Newton’s 2nd Law (sometimes referred to as conservation of momentum or balance of momentum)

$$F = ma,$$

i.e., force equals mass times acceleration. The left-hand side in (9) comes from ma , and the right-hand side denotes all the forces acting on the fluid. These consist of volume (body) forces and surface forces (or long-range and short-range forces). Volume forces act on all elements of the volume of a continuum. An example of a volume force is gravity. Volume forces are usually denoted by a vector valued function f as in equation (9). We note that in equation (8) we do not have the function f for simplicity.

The surface forces are what produces the deformation tensor, which in turn gives the Laplacian. First, the surface force acts on a surface element to which we assume we can assign a unit normal n . Then, the surface force, as a force is also a vector, which means it can be written in components. The i ’th component can be shown to be $T_{ij}n_j$, where T is the stress tensor (see for example [Bat99]). So if we consider a part of a fluid with volume V and enclosed by a surface S , the total surface force acting on S is given by

$$\int_S T_{ij}n_j dS = \int_V \partial_j T_{ij} dV,$$

where we used the divergence theorem. This is how we get (9) for fluids with constant density.

The question remains: what constitutes the stress tensor T ? For fluids at rest or also for perfect fluids, only normal stresses are exerted, and we have $T_{ij} = -p\delta_{ij}$, where p is the pressure, and gives ∇p in the equation. For fluids in motion or nonperfect fluids, we also have tangential stresses, which for isotropic fluids give an additional term in T_{ij} , which can be written as [Bat99, p.144]

$$2\mu D_{ij} - \frac{2}{3}\mu \operatorname{div} v \delta_{ij}, \quad (10)$$

where $D_{ij} = (\operatorname{Def} v)_{ij}$ is the deformation tensor, $\frac{1}{2}(\partial_j v_i + \partial_i v_j)$, and μ is the viscosity coefficient. Then, for an incompressible fluid, we are left with

$$\mu(\partial_j v_i + \partial_i v_j),$$

and after we take divergence as in (9) we get exactly

$$\mu \Delta v.$$

Viscosity is an internal friction of the fluid. In the case of the Euler equations, the fluid is assumed to be inviscid, to have zero viscosity. (Inviscid property, while an approximation, is also another good approximation in certain situations.) For inviscid fluids, there is no Laplacian in the equation, which also means there is no issue what the equation should be on the manifold. Another reason why it is clear what the Euler equations should be on general Riemannian manifolds is due to the existence of the variational principle, which we will briefly explain a little later.

Ebin and Marsden [EM70], when they introduce the deformation Laplacian, refer to Serrin [Ser59]. There, following Stokes, it is assumed that the stress-tensor T satisfies [Ser59, Section 59]:

1. T is a continuous function of the deformation tensor Def , and is independent of all other kinematic quantities.
2. T does not depend explicitly on the spatial position (spatial homogeneity).
3. There is no preferred direction in space (isotropy).
4. When $\operatorname{Def} = 0$, T reduces to $-pI$, where p is the pressure and I the identity matrix.

In case of a general manifold, it is not clear how one would assure properties 2 and 3. In fact, when Serrin discusses the Navier–Stokes equations in curvilinear coordinates (Section 13), Serrin remarks that, on a general manifold, it is not “evident how to formulate the principle of conservation of momentum.” However, right after, he says that “there seems to be no valid objection to taking Eq. (12.3) as a postulate.”

Equation (12.3) involves the divergence of the stress tensor, and it gives the deformation Laplacian if T satisfies 1 and 4 above and the remaining assumptions used by Serrin.

This connection with continuum mechanics was also pointed out by Taylor in [Tay92]. The connection does seem very natural indeed.

Arguments leading to the appearance of the deformation tensor are based on symmetry, and this is another reason why the deformation Laplacian might be the natural choice, as it involves the symmetric part of the covariant derivative. The Hodge Laplacian, for divergence-free vector fields, reduces to d^*d , and d corresponds to the antisymmetric part, and the Bochner Laplacian includes both the symmetric and the antisymmetric pieces.

While these three operators are types of Laplacians on a manifold, the above discussion illustrates that the stress-tensor is what encodes the effects of viscosity, and not the Laplacian. As a result, we should be really searching for the appropriate viscosity operator and not just the appropriate Laplacian.

Analysis: Argument Against the Hodge Laplacian on $\mathbb{H}^2$

With Chan and Disconzi [CCD17] we gave a mathematical argument coming from analysis of PDE that was against using the Hodge Laplacian on the hyperbolic plane, $\mathbb{H}^2$.

We showed that if we are working with the Hodge Laplacian, for a forced nonstationary equation on $\mathbb{H}^2$, one cannot establish an energy inequality as part of what is called a weak formulation of the equation. An energy inequality is an important tool in the existence theory of the equations, so the lack of such tool suggests that the Hodge Laplacian might not be the right operator for the Navier–Stokes equation on the hyperbolic plane.

We remark here that since the hyperbolic plane is non-compact, this result does not contradict what was done in [EM70] for compact manifolds with the Hodge Laplacian. Also, the weak formulation assumes data is in L^2 and seeks to find weak solutions that exist for all time. In [EM70], the initial data belongs to H^s for $s > \frac{n}{2} + 5$, and the solution is shown to exist for a short time.

Finally, the counterexample in [CCD17] would not apply to the use of the Hodge Laplacian on the sphere S^2 . This has to do with the equivalence of the L^2 norm of the covariant derivative and the exterior derivative d , $\|\nabla u\|_{L^2(S^2)}$ and $\|du\|_{L^2(S^2)}$, respectively. One could try to modify the norm in the case of $\mathbb{H}^2$, but there is no obvious modification (either it is not a norm or the same counterexample would apply).

Physics: Nonrelativistic Limit

Another way we can investigate the form of the Navier–Stokes equations on Riemannian manifolds is to consider the nonrelativistic limit of the relativistic Navier–Stokes equations. Nevertheless, in the case of relativistic fluids the “correct” form of the equations is not known either.

There are several theories in the literature, and at the time of writing [CCD17], we showed that they all could lead to the same equations where the Laplacian is the deformation Laplacian.

The main idea of the nonrelativistic limit is to start with the relativistic equation, and take the limit by assuming that the fluid velocities are very small compared to the speed of light. This means that we neglect terms of the order $\frac{|v|}{c}$, where v is the velocity of fluid particles and c the speed of light.

In this case, however, differently than the usual non-relativistic limit in the general theory of relativity, we considered the metric on $\{t = \text{constant}\}$ hypersurfaces converging to an arbitrary Riemannian metric. Usually, it is considered that the metric converges to the Minkowski metric, so the metric induced on $\{t = \text{constant}\}$ hypersurfaces is the Euclidean one. The manipulations we performed were formal, in a sense, we did not discuss the topology with respect to which the convergence was supposed to occur.

The fact that we arrived at the deformation Laplacian, in a certain sense, might not be a surprise. The relativistic equations are obtained as

$$\nabla^\alpha T_{\alpha\beta} = 0, \quad (11)$$

($\alpha, \beta = 0, \dots, 3$), where $T_{\alpha\beta}$ is a symmetric two-tensor. If $T_{\alpha\beta}$ depends on first order derivatives of the velocity, it will in general contain the term $\nabla_\alpha u_\beta + \nabla_\beta u_\alpha$, where u is the fluid’s four-velocity (which is the appropriate relativistic notion of velocity). This is indeed the case in all the relativistic theories we considered. Then, by taking the non-relativistic limit, it can be showed that the term $\nabla_\alpha u_\beta + \nabla_\beta u_\alpha$ produces $\nabla^i(\nabla_i v_j + \nabla_j v_i)$, where v is the (classical) fluid’s velocity ($i, j = 1, 2, 3$). This term gives the deformation Laplacian.

Since [CCD17], Disconzi has been involved in the development of new relativistic theories, which produced what is referred to as the BDNK model, named after Bemfica, Disconzi, Noronha, and Kovtun (see, e.g., [BDN22, Kov19] and references therein). Very recently, Hegade K R, Ripley, and Yunes studied nonrelativistic limit of the BDNK model [HKRRY23], and showed that the limiting equations (on the Minkowski space) depend on the choice of the parameters in the BDNK energy-momentum tensor. This would presumably carry over to Riemannian manifolds, and seems to imply that the nonrelativistic limit might not have the definitive predictive capability or at least, that this approach is more complicated, and more information about the problem the limiting equation is supposed to model might be needed.

Geometry: Gauss Formulas

Another way to look at this problem is to recall the following question: How can directional derivatives be defined on a submanifold embedded in a Euclidean space? One answer is: by taking the extrinsic directional derivative, and then projecting back to the submanifold. What is amazing is that this projected quantity coincides exactly with the intrinsic directional derivative. Moreover, the relationship between the intrinsic and the extrinsic quantities can be observed through the Gauss formula. We make this precise now.

Let M be an embedded submanifold in a Riemannian manifold $(\tilde{M}, \tilde{g})$, with the inclusion map, $\iota : M \hookrightarrow \tilde{M}$. Then the embedding induces a metric g on M , by $g = \iota^* \tilde{g}$. For example, in the case of the sphere S^2 embedded in $\mathbb{R}^3$, the induced metric on S^2 is just the Euclidean metric restricted to the vectors tangent to S^2 , and this is the standard, the so-called round metric on the sphere.

Now, if X, Y are vector fields on M , and $p \in M$, we can consider local extensions of X, Y to a neighborhood of p in $\tilde{M}$, and still denote them by X, Y . This is done so we can apply the extrinsic Levi-Civita connection on $\tilde{M}$, $\tilde{\nabla}$, to X, Y and evaluate it at $p \in M$, i.e, we consider

$$\tilde{\nabla}_X Y|_p, \quad p \in M.$$

By the definition of a connection, the result is a vector in $T_p \tilde{M}$. We can then decompose it into two parts: a part tangential to $T_p M$, denoted by $(\tilde{\nabla}_X Y)^T$, and a part normal to $T_p M$, denoted by $(\tilde{\nabla}_X Y)^\perp$. The normal part defines what is called the second fundamental form $\Pi : \mathfrak{X}(M) \times \mathfrak{X}(M) \rightarrow \Gamma(NM)$, where $\mathfrak{X}(M)$ denotes the smooth vector fields on M , and $\Gamma(NM)$ denotes the smooth sections of the normal bundle of M ,

$$\Pi(X, Y) = (\tilde{\nabla}_X Y)^\perp.$$

The second fundamental form has the nice properties that it is symmetric in X, Y , bilinear over smooth functions on M , and perhaps the most interesting property is that it is independent of the extensions of X, Y . If ∇ denotes the Levi-Civita connection on M corresponding to the induced metric g , then the Gauss formula tells us that the tangential part is $\nabla_X Y$, so we have

$$\tilde{\nabla}_X Y = \nabla_X Y + \Pi(X, Y), \quad (12)$$

when evaluated at $p \in M$.

There is a similar relationship between the intrinsic and extrinsic curvatures. We have

$$\begin{aligned} \tilde{R}(W, X, Y, Z) &= R(W, X, Y, Z) \\ &\quad - \tilde{g}(\Pi(W, Z), \Pi(X, Y)) \\ &\quad + \tilde{g}(\Pi(W, Y), \Pi(X, Z)), \end{aligned} \quad (13)$$

where $\tilde{R}, R$ are the Riemann curvature tensors on $\tilde{M}$ and M , respectively, and $\tilde{g}$ is the Riemannian metric on $\tilde{M}$.

To go back to the Navier–Stokes equation, when we discussed in the beginning taking each term in the equation and asking how it would look on a manifold, we can make the following observation. By Nash’s embedding, each Riemannian manifold can be embedded in $\mathbb{R}^m$ for some m . Then we could take the equation on $\mathbb{R}^m$ and project it onto the tangent space of the manifold in question. The projection is linear so we could apply it to each term individually. This process coincides with what we expected to get before for the partial in time of the velocity, the gradient of the pressure, and the nonlinear term: for the partial in time and the gradient, this is just a projection, and for the nonlinear term, we can apply the Gauss formula (12). What is left is again the question of the Laplacian. All the Laplacians related in (7) agree on the Euclidean space, so question can be reduced to: is there a Gauss formula for the Laplacian?

We looked into this initially with Chan and Disconzi for the sphere S^2 in [CCD17] with considering only the tangential part, and not trying to derive a general Gauss formula that includes both tangential and normal pieces. Still, we were excited to arrive at the tangential part being the deformation Laplacian. Later with Chan and Yoneda we wanted to extend this to the case of the ellipsoid. At first, we were surprised that the computations were a lot more involved, and later we realized that the general problem is more involved. In fact, the formula that we eventually derived in [CCY23] showed that the operator obtained by this procedure does depend on the extension of the vector field considered.

For example, in [CCD17] we worked with a vector field in spherical coordinates, and considered the simplest extension/restriction by not having the coordinate functions depend on the radial variable. That extension can be viewed as an extension so its norm grows depending on the distance away from the sphere. If we work with a vector field that has a norm preserved, then one could obtain the Hodge Laplacian instead! For the case of the sphere, this was actually already pointed out in [Ors74, Yam18].

In [CCY23], we extended the prior work to an ellipsoid, and we found a general formula that explicitly depends on the ambient/extrinsic vector field. It also generalizes the work of [CCD17, Ors74, Yam18] as it can be simplified to a formula that covers the case of the sphere.

In [CC22] with Chan, we extended the formula for the projected Laplacian to a general hypersurface embedded in an ambient manifold $\tilde{M}$, and also obtained the analog of the Gauss formula. For simplicity, we present the formula for a Euclidean hypersurface.

Theorem 1. [CC22] *Let $n \geq 2$; let M denote an embedded hypersurface in $\mathbb{R}^{n+1}$, $M \hookrightarrow \mathbb{R}^{n+1}$, $\dim M = n$; let v be a vector field on M , $p \in M$; and let $\tilde{v}$ be an extension of v to a*

neighborhood of p in $\mathbb{R}^{n+1}$, still denoted by v . Then at p

$$\begin{aligned} -\Delta v &= -\operatorname{div} \nabla v - \operatorname{Ric} v + nH[N, v] \\ &\quad - \tilde{\nabla}_N \tilde{\nabla}_N v + \tilde{\nabla}_{\tilde{\nabla}_N N} v \\ &\quad - \sum_i (2\kappa^i \nabla_i v^i + v(\kappa^i)) N, \end{aligned} \quad (14)$$

where

- Δv is the Euclidean Laplacian,
- $-\operatorname{div} \nabla v$ is the Bochner Laplacian on M ,
- $\operatorname{Ric} v = (\operatorname{Ric}(v, \cdot))^\sharp$, $\operatorname{Ric}(\cdot, \cdot)$ is the Ricci tensor on M ,
- H is the mean curvature of M ,
- N is the choice of the unit normal on M ,
- $[\cdot, \cdot]$ is the Lie bracket,
- $\tilde{\nabla}$ is the Euclidean connection,
- κ^j are the principal curvatures of $M \hookrightarrow \mathbb{R}^{n+1}$, and
- $\kappa^i \nabla_i v^i$ is the divergence of v weighted by the principal curvatures.

Hence, if we are only interested in the projected piece, we arrive at

$$\begin{aligned} (-\Delta v)^T &= -\operatorname{div} \nabla v - \operatorname{Ric} v + nH[N, v]^T \\ &\quad - (\tilde{\nabla}_N \tilde{\nabla}_N v)^T + (\tilde{\nabla}_{\tilde{\nabla}_N N} v)^T. \end{aligned} \quad (15)$$

We discuss this formula for the case of the sphere.

We can let $N = \partial_\rho$, where ρ is the radial variable in the spherical coordinates, and work with an orthonormal frame obtained from the spherical coordinates

$$E_1 = \frac{\partial_\phi}{\rho}, \quad E_2 = \frac{\partial_\theta}{\rho \sin \phi}, \quad E_3 = N = \partial_\rho.$$

We then let $v = v^\alpha E_\alpha$, where we sum α from 1 to 3, and compute

$$\tilde{\nabla}_N N = \tilde{\nabla}_{\partial_\rho} \partial_\rho = 0 \quad \text{and} \quad \tilde{\nabla}_N v = \partial_\rho(v^\alpha) E_\alpha.$$

So

$$\tilde{\nabla}_N \tilde{\nabla}_N v = \tilde{\nabla}_N (\partial_\rho(v^\alpha) E_\alpha) = \partial_\rho^2 v^\alpha E_\alpha.$$

Also

$$\operatorname{Ric} v = v,$$

$$nH = -2, \quad \text{and} \quad [N, v] = \tilde{\nabla}_N v - \tilde{\nabla}_v N = \partial_\rho v^\alpha E_\alpha - v,$$

where we use in the last line the shape operator $sv := -\tilde{\nabla}_v N$, which is simply $sv = -v$ in the case of the sphere.

Projecting and using formula (15), we have

$$(\tilde{\nabla}^* \tilde{\nabla} v)^T = \nabla^* \nabla v + v - 2\partial_\rho v^i E_i - \partial_\rho^2 v^i E_i, \quad (16)$$

where now the Roman indices i sum from 1 to 2, and E_i is an ON frame on the sphere.

If we wish to arrive at the deformation Laplacian, by (7) we would need

$$v - 2\partial_\rho v^i E_i - \partial_\rho^2 v^i E_i = -v.$$

A computation shows that if $v = v^i E_i$ on S^2 , and in general on $\mathbb{R}^3$, $v = \rho v^i E_i$, then we do get $-v$. On the other hand, if v is always $v = v^i E_i$, then we do get the Hodge Laplacian.

These are the examples we mentioned above. We take an in-depth look at these in an upcoming article with Chan and Fuster.

A reader might be wondering here: but we are looking for solutions of the Navier–Stokes equation, how do we know that these above vector fields solve the Euclidean Navier–Stokes equations? This is a good point, and we look at relating the solutions of the Euclidean Navier–Stokes equations to the ones on an embedded submanifold in the next section.

Analysis and Physics: Thin Shell Limit

Consider the unit sphere, S^2 , embedded in $\mathbb{R}^3$, and a thin shell around it of 2ϵ thickness, where ϵ is small. This is a 3D domain with boundary. The boundary has two components, the sphere of radius $1 - \epsilon$ and the sphere with radius $1 + \epsilon$. We could study the Euclidean Navier–Stokes equations in this domain, or a domain where one boundary is the sphere itself and the other, for example, the sphere of radius $1 + \epsilon$, and then take the limit of the solutions when $\epsilon \rightarrow 0^+$.

Such procedures have been performed in elasticity, and also in the case of fluids. For example, in [TZ97], Temam and Ziane take the average of the Euclidean Navier–Stokes solutions and show they converge to the solution of the Navier–Stokes on S^2 , where the Laplacian is the Hodge Laplacian. In [Miu20], Miura considers a more general domain for a more general surface and takes the limit, and arrives at the Navier–Stokes equation on the surface where the Laplacian is the deformation Laplacian. How can this be?

The answer lies in the choice of the boundary conditions. The thin shell has a boundary, and to solve the Navier–Stokes equation, one needs to impose a boundary condition. Temam and Ziane work with the boundary condition, where it is assumed that u is tangential on the boundary, and where the cross product of $\operatorname{curl} u$ with the normal is zero. This is sometimes called the Hodge condition. Miura, on the other hand, uses the Navier’s slip condition, which in its simplest form can be written as

$$((\operatorname{Def} u)_{ij} n)^T = 0,$$

where we think of the deformation tensor now as the matrix, and we apply it to the normal vector, and then project it onto the boundary.

In an upcoming article with Chan and Fuster, we show that the resulting equation depends not just on the boundary conditions, but also on how we take the average.

Analysis and Geometry: Asymptotic Behavior of the Solutions

Interesting observations have been made by Samavaki and Tuomela [ST20], where the authors consider the equations on compact manifolds. There it is shown that the

difference in operators does make a difference even on compact manifolds, and even in a simple case of the sphere, S^2 .

The authors investigate qualitative behavior of the solutions for the time-dependent (nonstationary) problem. In particular, the importance of the Killing fields is highlighted. It is shown that for the solutions with the deformation Laplacian, the solution will converge to a Killing field as $t \rightarrow \infty$. On the other hand, if the Hodge Laplacian is used, the solutions will tend to a harmonic form. Therefore, as it is pointed out, in the case of the sphere, where there are no nontrivial harmonic forms, the solutions for the equation with the Hodge Laplacian will converge to zero, whereas the solutions with the deformation Laplacian will converge to a Killing field on the sphere, which corresponds to rotations.

Probability: Stochastic Action Principle and Lagrangian Paths

The Navier–Stokes equations on compact manifolds with the Hodge Laplacian and the deformation Laplacian have appeared in probabilistic approaches.

In the work of Arnaudon and Cruzeiro [AC12], the Navier–Stokes equation with the Hodge Laplacian is recovered in the connection with a variational principle for stochastic Lagrangian flows. Then Arnaudon, Cruzeiro, and Fang [ACF18] connected the variational principle with the Navier–Stokes equation with the deformation Laplacian.

To put this in context, the seminal work of Arnold [Arn66] and Ebin and Marsden [EM70] shows how the solutions to the Euler equation can be realized as minimizers that are geodesics on the infinite-dimensional group of the volume preserving diffeomorphisms. Arnold observed that the group of the volume preserving diffeomorphisms is a Lie group that can be endowed with L^2 Riemannian metric, which is the kinetic energy. This approach takes on what is called the Lagrangian point of view. This means we are interested in tracking individual particles and their position. Until now, we were interested in tracking the velocity of the fluid at each point in space, which is called the Eulerian point of view.

In the Lagrangian point of view, the positions are being tracked by the particle paths γ satisfying

$$\partial_t \gamma(t, x) = u(t, \gamma(t, x)), \quad \gamma(0, x) = x,$$

where u is the velocity field and satisfies the Euler equation.

The action that the particle paths are minimizing is given by

$$A(\gamma) = \frac{1}{2} \int_0^T \int_M g(\partial_t \gamma, \partial_t \gamma) \, dx \, dt.$$

This approach is called often the variational principle or the least action principle. It is not clear how to extend

this principle to the Navier–Stokes equations. The work in [AC12] and [ACF18] establishes the principle in the stochastic setting, but what is important to point out is that the word stochastic only refers to the Lagrangian paths: the velocity solves the deterministic Navier–Stokes equations.

A related approach, also using stochastic Lagrangian paths for the Navier–Stokes equations, is based on the results of Constantin and Iyer [CI08], and it was presented for general compact manifolds by Fang [Fan20]. Here, the main idea of this approach can be related to the Feynman–Kac formula, which connects solutions of linear parabolic PDEs with a stochastic process, which solves a corresponding stochastic differential equation. Constantin and Iyer extend this to the nonlinear system of the Navier–Stokes equations in the Euclidean setting, and [Fan20] establishes the representation formula for the Navier–Stokes equation on general compact manifold with the Hodge Laplacian.

Summary

We have discussed different approaches one can consider when looking at the question of what the Navier–Stokes equations should look like on a general Riemannian manifold. As interesting as all the different approaches are, none of them currently seems to lead to a definite answer. For a while, it was this author’s personal bias that the continuum mechanics argument was the most convincing one. At this point, we believe that the exact form of the equations will depend on the physical problem at hand.

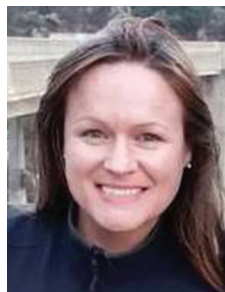
The methods we would like to explore further are the methods related to Gauss formulas as well as the original derivations of the Navier–Stokes equations in the Euclidean setting. The Euclidean equation has an interesting history itself, as it was derived at least five times, starting with Navier, Cauchy, Poisson, Saint-Venant, and Stokes. Revisiting these derivations might lead to interesting insights.

We end by saying that whichever model we choose to work on, ideally we would like to be able to justify it with some method. At the same time, as pure mathematicians, we could study any problem we find interesting, whether for its physical or mathematical motivation.

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